

Unit 1.2 Absolute value p A6.

Note Title

2014/01/27

Distance between a and 0 denoted by $|a|$ always positive.

$$|-3| = 3 \quad |4| = 4$$

Definition p A6

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$|x-2| = \begin{cases} x-2 & \text{if } x-2 \geq 0 \\ -(x-2) & \text{if } x-2 < 0 \end{cases}$$

Piece wise defined function.

$$= \begin{cases} x-2 & \text{if } x \geq 2 \\ -x+2 & \text{if } x < 2 \end{cases}$$

$$|2x+1| = \begin{cases} 2x+1 & \text{if } 2x+1 \geq 0 \\ -(2x+1) & \text{if } 2x+1 < 0 \end{cases}$$

$$= \begin{cases} 2x+1 & \text{if } x \geq -\frac{1}{2} \\ -2x-1 & \text{if } x < -\frac{1}{2} \end{cases}$$

$$\boxed{4.} \quad \sqrt{x^2} = |x|$$

$$\sqrt{(-2)^2} = |-2| = 2$$

$$|-3 \cdot 6| = |-18| = 18$$

$\boxed{5}$ Properties p A7

$$a, b \in \mathbb{R} \quad \text{and} \quad n \in \mathbb{Z}$$

$$|-3||6| = 3 \cdot 6 = 18$$

$$\textcircled{1} \quad |ab| = |a||b|$$

$$\textcircled{2} \quad \left| \frac{a}{b} \right| = \frac{|a|}{|b|}$$

$$\textcircled{3} \quad |a^n| = |a|^n$$

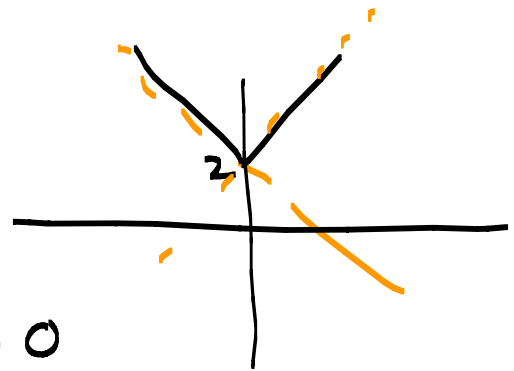
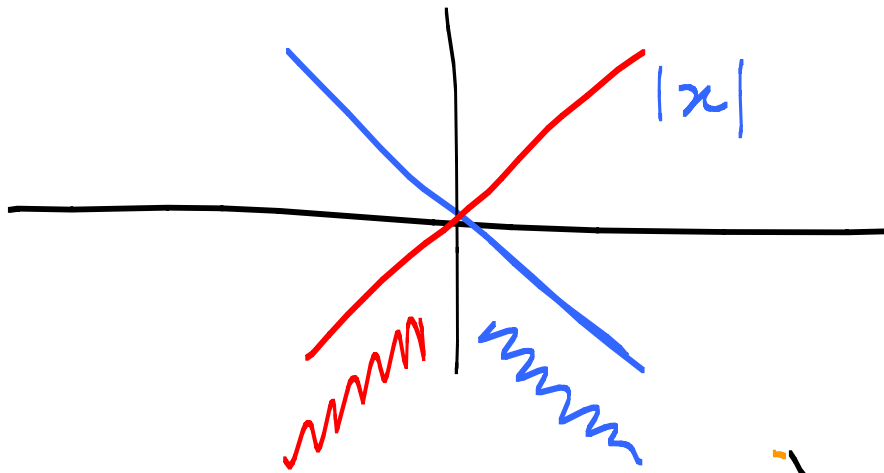
$\boxed{\text{Add}}$

$$\textcircled{4} \quad |x| \geq 0 \quad \text{for all } x \in \mathbb{R} !$$

$$\textcircled{5} \quad |x| = 0 \quad \text{if and only if } x = 0$$

What does $y = |x|$ look like?

$$y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

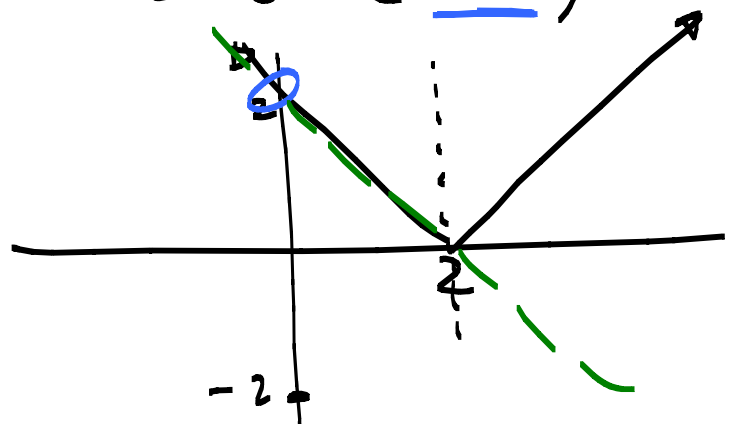


$$y = |x| + 2 = \begin{cases} x + 2 & \text{if } x \geq 0 \\ -x + 2 & \text{if } x < 0 \end{cases}$$

$$y = |x - 2| \rightarrow \text{move 2 units} \rightarrow$$

$$= \begin{cases} x - 2 & \text{if } x - 2 \geq 0 \quad (x \geq 2) \\ -\underline{(x - 2)} & \text{if } x - 2 < 0 \quad (\underline{x} < 2) \end{cases}$$

$\rightarrow -x + 2$



$$y = |\underbrace{x^2 - 1}|$$

$$= \begin{cases} \underline{\underline{x^2 - 1}} & \text{if } x^2 - 1 \geq 0 \\ -(x^2 - 1) & \text{if } x^2 - 1 < 0 \end{cases}$$

if

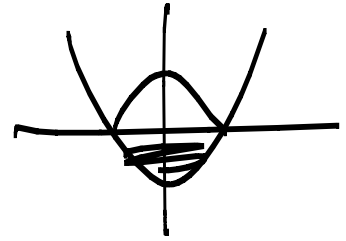
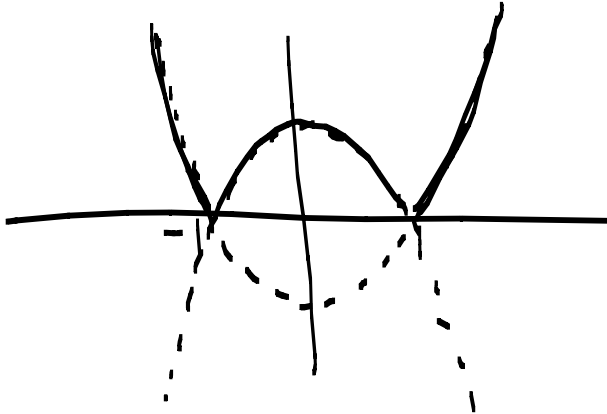
$$x^2 - 1 \geq 0$$

or $x \in (-\infty, -1] \cup [1, \infty)$

if

$$x^2 - 1 < 0$$

or $x \in \underline{\underline{(-1, 1)}}$



Solving equations or inequalities. pA7

6. Important. Suppose $a > 0$, then

$$4. |x| = a \text{ if and only if } x = \pm a$$

$$|-3| = 3$$

$$|3| = 3$$

$$|x| = 3$$

$$\Rightarrow x = \pm 3$$

① Solve x if $|x+9| = 4$

$$\Rightarrow x+9 = \pm 4$$

$$\Rightarrow x = \pm 4 - 9$$

$$= -5 \text{ or } -13$$

② Solve $|x^2 - 9| = 14$

$$\Rightarrow x^2 - 9 = \pm 14$$

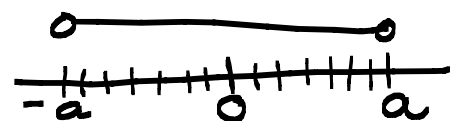
$$\Rightarrow x^2 = 23 \quad \text{or} \quad x^2 = -5$$

$$\Rightarrow x = \pm \sqrt{23}$$

$$x \notin \mathbb{R}$$

5. $|x| < a$ if and only if $-a < x < a$

↑



① $|x| < 4$

$\Rightarrow -4 < x < 4$

② $|x+4| \leq 4$

$\Rightarrow -4 \leq x+4 \leq 4$

$\Rightarrow -8 \leq x \leq 0$

$x \in [-8, 0]$

6. $|x| > a$ if and only if $x > a$ or $x < -a$



① $|x| > 6$

$\Rightarrow x > 6$ or $x < -6$

$x \in (-\infty, -6) \cup (6, \infty)$

② $|2x-3| \geq 6$

$\Rightarrow 2x-3 \geq 6$

$\Rightarrow x \geq \frac{9}{2}$

or $2x-3 \leq -6$

$x \leq -\frac{3}{2}$

$x \in (-\infty, -\frac{3}{2}] \cup [\frac{9}{2}, \infty)$

Ex

① Solve $\left| \frac{2x-1}{x+1} \right| = 3 \quad (x \neq -1)$

$$\Rightarrow \frac{2x-1}{x+1} = 3 \quad \text{or} \quad \frac{2x-1}{x+1} = -3$$

$$\Rightarrow 2x-1 = 3x+3$$

$$\Rightarrow x = -4$$

$$2x-1 = -3x-3$$

$$5x = -2$$

$$x = -\frac{2}{5}$$

② $|2x-4| > 8$ Ans: $x \in (-\infty, -2) \cup (6, \infty)$

③ $|x-3| \leq 6$ Ans: $x \in [-3, 9]$

④ $|x+3| = |2x+1|$ $|a| = \sqrt{a^2}$
 $|x+3|^2 = |2x+1|^2$ $\Rightarrow |a|^2 = a^2$
 $(x+3)^2 = (2x+1)^2$

$$x^2 + 6x + 9 = 4x^2 + 4x + 1$$

$$\Rightarrow 3x^2 - 2x - 8 = 0$$

$$\Rightarrow (3x+4)(x-2) = 0$$

$$\Rightarrow x = -\frac{4}{3} \quad \text{or} \quad x = 2$$

⑤ $\frac{x}{|x-2|} > 0$

$$\Rightarrow x > 0 \quad \text{since}$$

$$|x-2| \geq 0, \text{ but}$$

$$\frac{x}{x-2} > 0 ?$$

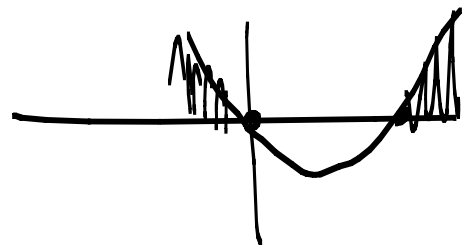
$$(x-2)^2 \times \frac{x}{x-2} > (x-2)^2 \cdot 0$$

$$(x-2)x > 0$$

$$x \neq 2$$

$$x \in (0, \infty) \setminus \{2\}$$

$$x \in (0, 2) \cup (2, \infty)$$



$$x \in (-\infty, 0) \cup (2, \infty)$$

$$(6) \frac{|x|}{|x-1|} \geq 0$$

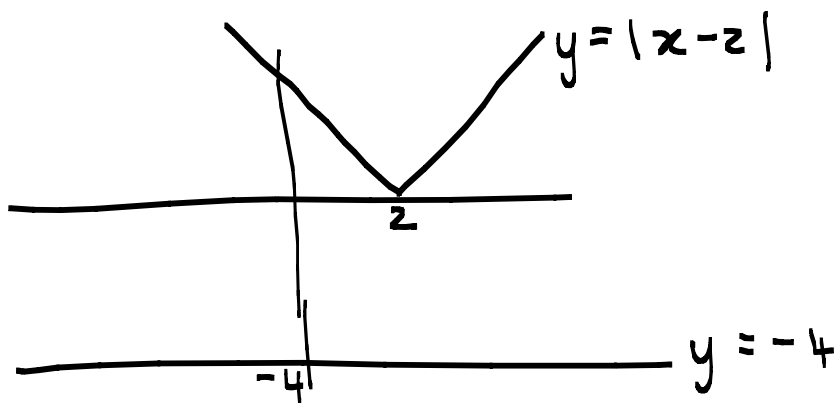
$$x \in \mathbb{R} \setminus \{1\}$$

$$x \in (-\infty, 1) \cup (1, \infty)$$

$$(7) |x-2| < -4$$

$x \in \emptyset$. No solution.

(8) Graph $y = |x-2|$ and show that $|x-2| < -4$ is not possible.



$$y = |x|$$

$$y = -4$$

$$x \in \{ \} \quad \text{or} \quad x \in \emptyset$$